

PROJECT ON DATA SCIENCE AT WORK
TOPIC- BUSINESS INSIGHTS ON “HOW TO BETTER
CLINICAL OUTCOMES OF IVF TREATMENT”

SUBMITTED BY

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BATCH- DSBA-NOV’20

DATE: 25th SEP 2021

I am a Healthcare Professional with over a decade of experience in Lifestyle Medicine, Healthcare Operations and Business Development. My work experience spans hospitals and organizations at various stages of growth from Administrative roles in Multi-specialty hospitals to P & L Head role in Single specialty IVF & Mother & Child Health Business. My learning inclination has always driven me to explore new areas and this is what led me to do the course in Data Analytics.

I am very passionate about Healthcare and Technology and want to apply the science of Data Analytics in deriving useful insights in healthcare sector. I have been handling P&L for Healthcare Business Units for Infertility and Elder Care specialties. My current project for Data Science at Work is regarding the dataset for IVF Clinical Protocol used to perform IVF –ICSI on patients with resulting Beta HCG Result of the patients. Hospitals majorly face the problem of lack of technology and maintenance of databases. This makes it very difficult to derive reliable insights during analysis.

I have tried to apply my learnings at Great Learnings by performing EDA and Visualization to derive insights on the resulting Clinical Outcome on the ICSI procedure

Problem Statement and the Approach

IVF or In Vitro Fertilization is a medical procedure to treat infertility in patients. The dataset has been collected for patients over the years 2020-2021 who have undergone the ICSI procedure. The dataset has 205 rows and 18 columns with various attributes like clinical protocol, medication used for the ICSI procedure with resulting endometrium thickness, no of oocytes retrieved, no of embryos formed and the clinical outcome which is the Beta HCG Result.

I have used Jupyter notebook to import the required python libraries, read the dataset using pandas read function. Using the EDA method the dataset was pre-processed before moving forward for univariate, bivariate and multivariate analysis.

Recommendation for Adopting Best Clinical Practise

1. Though there are three protocols and most of the patients have undergone Antagonist Protocol, patients using the Down Regulation Protocol have shown better positive results at 63%
2. Embryo Transfer Type Frozen Embryo transfer has fared better outcome at 50% positive pregnancies
3. The median value of eggs retrieved through different protocols is highest with Down Regulation type at 16 eggs
4. The median no of embryos formed using different protocols is similar for DR and Antagonist but higher upper quartile with DR. The median embryos formed is 5
5. In terms of using different GNRH, Aspirelix and Cetrotide have fared better clinical outcomes
6. Multicollinearity is present between Eggs Retrieved and Embryos formed, mature eggs and embryos formed and embryos formed and freeze
7. Negative linear relationship is present between Age and Embryos formed with negative Beta HCG Result